

Introduction to Artificial Intelligence

Rational Agents

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What is an Intelligent Agent?

Think like humans	Think rationally	Thinking
Act like humans	Act rationally	Acting
Like a human	Rationally	

Rational Agents

What identifies an agent?

- it exists in a (constrained) task environment
- it has sensors to perceive the environment
- it has actuators to interact with the environment
- it has a task to perform

Rational Agents

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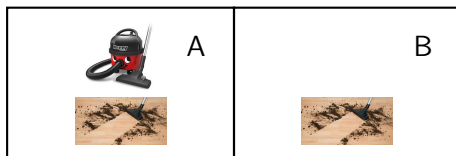
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Definition: Percept

A *percept* is the agent's information provided by a sensor.

The *percept sequence* is the sequence of percepts of the agent.

Automatic Vacuum Cleaner I



- percepts are pairs of (room label, room state)
- actions are goLeft, goRight, cleanUp, doNothing

Possible percepts: ???

Possible percept sequences: ???

The agent acts

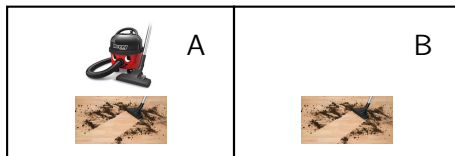
The actions of the agent are determined by the percept sequence.

Definition: Agent function

An *agent function* is a map from percept sequences to actions.

The *agent program* is an implementation of the agent function on a physical architecture.

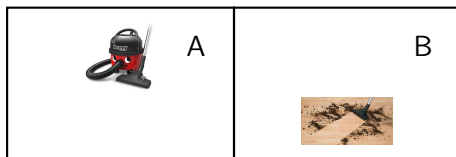
Automatic Vacuum Cleaner II



Agent function (seen as a table):

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⋮	⋮

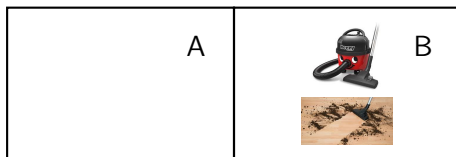
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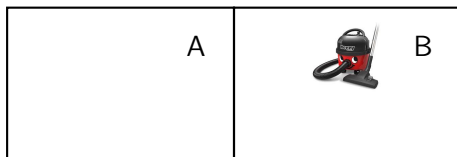
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Rational Agents

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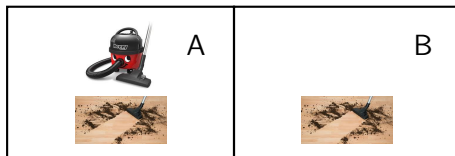
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Definition: Rational Agent

An agent is rational if it selects an action that is expected to maximise its performance measure for each possible percept sequence, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Automatic Vacuum Cleaner III



Percept sequence	Action	Performance measure: maximize "clean"-percepts
(A, clean)	goRight	
(A, dirty)	cleanUp	
(B, clean)	goLeft	
(B, dirty)	cleanUp	
(A, clean), (B, clean)	goLeft	
(A, dirty), (A, clean)	goRight	
⋮	⋮	

Possible variations of the agent function/performance measure

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- autonomy = the agent is capable of learn and not completely depend on the prior knowledge

Task Environment Specification

When designing an agent, we need to specify the task environment:

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- **Environment:** the world in which the agent operates

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- **Sensors:** how the agent perceives the environment

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- **Actuators:** how the agent can interact with the environment

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- **S**ensors: how the agent perceives the environment

The book this is called the **PEAS** description of the task environment.

Examples PEAS I

Agent

P

E

A

S

Vacuum cleaner

Taxi driver

Robotic Arm

- **P**erformance measure
- **E**nvironment
- **A**ctuators
- **S**ensors

Examples PEAS I

Agent	P	E	A	S
Vacuum cleaner	Maximize clean, maximize time rooms are clean	Rooms, dirt	Wheels, hose	Dirt sensor, location sensor

Taxi driver

Robotic Arm

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Agent	P	E	A	S
Vacuum cleaner	Maximize clean, maximize time rooms are clean	Rooms, dirt	Wheels, hose	Dirt sensor, location sensor
Taxi driver	Fast, friendly, profits	Roads, obstacles, customers	Accelerator, steering wheel, brake, horn	Cameras, sonar, speedometer, distance sensor

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Robotic Arm	Amount of items correctly placed	Conveyor belt, containers	Servo motors, robot arm, robot hand	Camera, arm and hand position sensors

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- Known vs unknown [knowledge about the environment dynamics]

Examples PEAS II

Task Environment	Observability	Many agents?	Deterministic?	Episodic	Dynamic?	Continuous?
Crossword puzzle						
Chess with clock						
Poker						
Vacuum Cleaner						
Taxi driving						
Medical diagnosis						
Robotic Arm						

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Agents Types

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⇒

```
Agent function:  
while ON do  
  if (room, dirty) then  
    clean up  
  else  
    if room = A then  
      go right  
    else  
      go left  
    end if  
  end if  
end while
```

Reflex Agents

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If you want to play with reflex agents, see [bugworks](#)

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Model-based Reflex Agents

- Store some information about the world [memory]
- Update the internal state based on percepts and the model
- Select actions based on the history of percepts
- Knowing the current state does not mean knowing the goal!

Goal-based Agents

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- Model-based agent with the ability to plan ahead and determine preference

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- E.g., Binary goal (happy / unhappy state)

Utility-based Agents

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- Goal-based agent with a utility function to evaluate different states

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- Goal-based agent with a utility function to evaluate different states
- E.g., Maximize percepts “clean” while minimizing the number “goLeft” and “goRight”

Learning Agents

- Can update their model of the world through learning:
 - Performance element: select actions using the current knowledge
 - Learning element: improvements based on feedback
 - Critic: provides feedback on the agent's actions to guide learning
 - Problem generator: suggests exploratory actions to discover new knowledge
- Learning = adapt the behavior of the agent based on “experience”
- Autonomy = the agent is capable of learning and not completely depend on the prior knowledge

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- Goal-based agents: model-based agent + goals but no notion of preferences;
- Utility-based agents: goal-based agent + utility function to evaluate different way to achieve the goal;
- Learning agents: utility-based agent + learning capabilities to update the model of the world;

Summary

- Definition of an agent and rationality
- PEAS description of task environments
- Properties of task environments
- Types of agents

Book references:

- Chapter 2

Next TA session:

- Exercises on PEAS and agents description

Next lecture:

- Exploring the (task) environment